Storyboard

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1. The Question

Define the purpose and scope of the analysis.

Main Question: How did COVID-19 affect public health across different demographics and regions?

Sub-questions:

- What are the trends in COVID-19 cases, hospitalizations, and deaths over time?
- How do these trends vary by age, sex, and geographic location?
- What role did vaccination rates play in reducing severe outcomes?

2. The Dataset

Summarize the data sources and key variables.

Dataset: COVID-19 case data from a government health agency or a trusted data repository.

Variables are shown below:

1. Date: Date of case reporting.
2. Region: Geographic location (e.g., country, state, county).
3. Age: Age group (e.g., 0–18, 19–64, 65+).
4. Sex: Biological sex.
5. Cases: Number of confirmed cases.
6. Deaths: Number of deaths.
7. Hospitalizations: Number of hospitalizations.
8. Vaccination Rate: Proportion of vaccinated individuals in a region.

3. Initial Explorations

Describe the exploratory analysis steps.

Missing Data Analysis: Check for and handle missing or inconsistent values.

Data Summaries: Calculate total cases, deaths, hospitalizations, and vaccination rates for key groups.

Visualizations: Time series plots for trends in cases and deaths. Bar charts showing demographic distributions of severe outcomes. Maps to visualize geographic variations.

Tools in R: Use dplyr for data wrangling. Use ggplot2 for visualizations. Use tidyr for handling missing data.

4. The Approach

Explain the planned analysis steps and methods.

Descriptive Analysis: Aggregate cases, deaths, and hospitalizations by region, age, and sex. Plot trends over time for key metrics.

Comparative Analysis: Compare regions with high and low vaccination rates. Test for significant differences in outcomes using statistical methods (e.g., t-tests, chi-squared tests).

Modeling: Build a linear or logistic regression model to assess the effect of vaccination rates on outcomes. Include control variables like region, age, and sex.

5. The Results

Summarize the key findings.

Case Trends: Regions with higher vaccination rates showed slower growth in cases.

Demographic Differences: Older age groups had significantly higher hospitalization and death rates.

Vaccination Impact: Vaccination rates were strongly associated with lower severe outcomes in the regression model.Include visualizations (e.g., annotated time series plots and regression coefficient plots) to highlight findings.

6. The Communication

Outline how the results will be communicated.

Deliverables:

- Report: A formal analysis report with visualizations and statistical results.
- Presentation: A slide deck summarizing the key findings and policy recommendations.
- Interactive Dashboard: Built using Shiny or similar tools to allow users to explore the data.

Narrative Structure:

- Introduction: Present the research questions and importance of the study.
- Methods: Describe the data sources and analysis approach.
- Results: Showcase findings using clear visualizations.
- Discussion: Discuss limitations, policy implications, and areas for further study.